

Part 3: SQL

1. Introduction to SQL

- Tables
- Field Naming
- Data Type
- Introduction to Query
 - Writing Queries
 - SQL Flavour

2. Intermediate SQL

- Querying a Database
- Query Execution
- Debugging SQL
- SQL Style
- Filtering Numbers
- Comparison Operators
- Filtering Text
- Aggregate Functions (*new*) — COUNT, SUM, AVG, MIN, MAX
- Summarizing Data
- Summarizing Subsets
- Aliasing and Arithmetic
- Sorting & Grouping Results
- NULL Handling (*new*)
- CASE Statements (*new*)

3. Fundamental SQL

- Joins
- Defining Relationships
- Multiple Joins
- UNION / UNION ALL (*new*)

4. Manipulation in SQL Statements

- Subqueries
- Order of Operation of Subqueries
- Correlated Subqueries
- Nested Subqueries
- Common Table Expressions (CTEs)
- The Reason for Using CTEs
- Difference Between Joins & Correlated Subqueries
- Multiple/Nested Subqueries
- Window Functions
- Sliding Window
- Ranking Functions
- Paging
- Frames
- Moving Average and Total
- Pivoting

5. Functions for Manipulating Data

- Common Data Types

- Data and Type — Data Type
- Working with Arrays
- Searching Arrays
- Retrieving the Current Timestamp
- Extracting and Transforming Date & Time Data
- Changing the Case of a String
- Parsing String and Character Data
- The Key to a Database
- Column Constraints
- Data Type
- Numeric Data Type and Summary Functions

6. Creating and Managing Tables

- Create Table
- Insert Into Table
- Update Table (*new*)
- Alter Table (*new*)
- Delete & Drop Table